ALM-5

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Eg:- Tabular & set form 0/1 knapsack problem

$P = \{5, 9, 10, 13\}$ $m = 12$

$W = \{5, 6, 7, 9\}$ $n = 4$

P	W	•	0	1	2	3	4	5	6	7	8	9	10	11	12
		0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	5	1	0	0	0	0	0	5	5	5	5	5	5	5	5
9	6	2	0	0	0	0	0	5	9	9	9	9	9	14	14
10	7	3	0	0	0	0	0	5	9	10	10	10	10	14	15
13	9	4	0	0	0	0	0	5	9	10	10	13	13	14	15

$V[i, w] = \max\{V[i-1, w], V[i-1, w-w_i] + p_i\}$

$V[4,1] = \max\{V[3,1], V[3, 1-9] + 13\} = 0$

$V[4,8] = \max\{V[3,8], V[3, 8-9] + 13\} = 0$

$V[4,9] = \max\{V[3,9], V[3,0] + 13\} = 13$

$V[4,10] = \max\{V[3,10], V[3,1] + 13\} = 13$

$V[4,11] = \max\{V[3,11], V[3,2] + 13\} = 14$

$V[4,12] = \max\{V[3,12], V[3,3] + 13\} = 15$

$V[3,4] = \max\{V[2,4], V[2, 4-7] + 10\} = 0$

$V[3,5] = \max\{V[2,5], V[2, 5-7] + 10\} = 5$

$V[3,7] = \max\{V[2,7], V[2, 0] + 10\} = 10$

$V[3,8] = \max\{V[2,8], V[2,1] + 10\} = 10$

$V[3,9] = 10$

$V[3,10] = \max\{V[2,10], V[2,3] + 10\} = 10$

$V[3,11] = \max\{V[2,11], V[2,4] + 10\} = 14$

$V[3,12] = 15$

$V[2,4] = \max\{V[1,4], V[1,4-6]+9\} = 0$

$V[2,5] = \max\{V[1,5], V[1,4]+9\} = 5$

$V[2,6] = \max\{V[1,6], V[1,0]+9\} = 9$

$V[2,10] = \max\{V[1,10], V[1,10-6]+9\} = 9$

$V[2,11] = \max\{V[1,11], V[1,11-6]+9\} = 14$

$V[2,12] = \max\{V[1,12], V[1,6]+9\} = 14$

	x_1	x_2	x_3	x_4	
	1	0	1	0	$15-0=15$
					$15-10=5$
					$5-5=0$

The maximum profit is obtained by taking x_1, x_3

The maximum profit is $5+10=15$

Set method

$P = \{5, 9, 10, 13\}$

$W = \{5, 6, 7, 9\}$

$m = 12$

$n = 4$

$S^0 = \{(0, 0)\}$

$S_1^0 = \{(5, 5)\}$

$S^1 = \{(0,0), (5,5)\}$

$S_1^1 = \{ \cancel{(5,5)}, (9, 6), (14, 11)\}$

$S^2 = \{(0,0)\ (5,5)\ (9,6)\ (14,11)\}$

$S_1^2 = \{(10, 7)\ (15, 12)\ (19, 13)\ (24, 18)\}$

$S^3 = \{(0,0)\ (5,5)\ (9,6)\ (14,11)\ (10,7)\ (15,12)\}$

$\cancel{(19,13)\ (24,18)\}}$

$S_1^3 = \{(13, 9), (18, 14), (22, 15), (27, 20)\ (23, 16), (25, 21)\}$

$S^4 = \{(0,0)\ (5,5)\ (9,6)\ (14,11)\ (10,7)\ (15,12)$

$(13, 9)\}$

$S^4 = \{(0,0)\ (5,5)\ (9,6)\ (10,7)\ (13,9)\ (14,11)$

$(15,12)\}$

(i) $(15, 12) \in S^4$

$(15, 12) \in S^3$ $\boxed{x_4 = 0}$ $(15-0, 12-0)$

$15, 12$

(ii) $(15, 12) \in S_3$

$(15, 12) \notin S_2$ $\boxed{x_3 = 1}$ $(15-10, 12-7)$

$(5, 5)$

(iii) $(5,5) \in S_2$

$5,5 \in S_1$ $\boxed{X_2 = 0}$

(iv) $(5,5) \in S_1$

$(5,5) \notin S_0$ $\boxed{X_1 = 1}$

The max profit is 15

~~Maxim~~

~~The maximum profit~~ is

ALM-6

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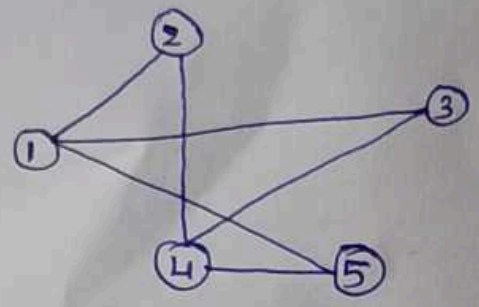
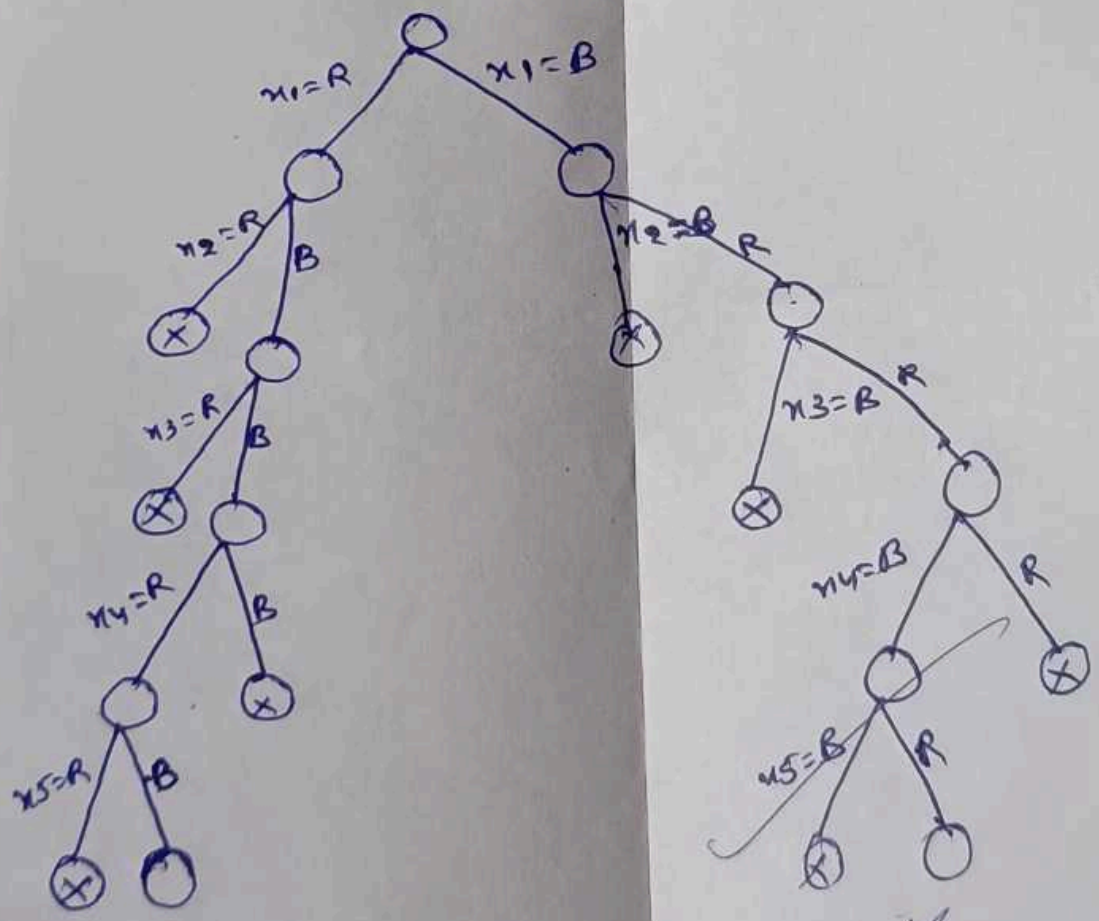
Design and Analysis of Algorithms

Graph colouring problem find chromatic number

1)



A) State space tree



The solutions are

[R, B, B, R, B]

[B, R, R, B, R]

Hence the chromatic number is 2.